

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2455604.690151 +/- 7.2e-05 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1317 +/- 0.0079
Transit depth (Rp/Rs)^2	1.73 +/- 0.21 [%]
Orbital Inclination (inc)	87.14 +/- 0.46
Airmass coefficient 1 (a1)	1.124 +/- 0.0085
Airmass coefficient 2 (a2)	-0.1059 +/- 0.0031
Scatter in the residuals of the lightcurve fit is	0.76 %
Best Comparison Star	None
Optimal Aperture	3.14
Optimal Annulus	8.47
Transit Duration (day)	0.0902 +/- 0.0063

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1317 $\pm$ 0.0079 vs 0.131 (deviation 0.09 σ)
Duration fit vs published	0.0902 d vs 0.1075 d (deviation 2.75 σ)
Mid-time O-C	-0.0 min
Depth SNR	16.7
Residual scatter	0.76 %

Light curve

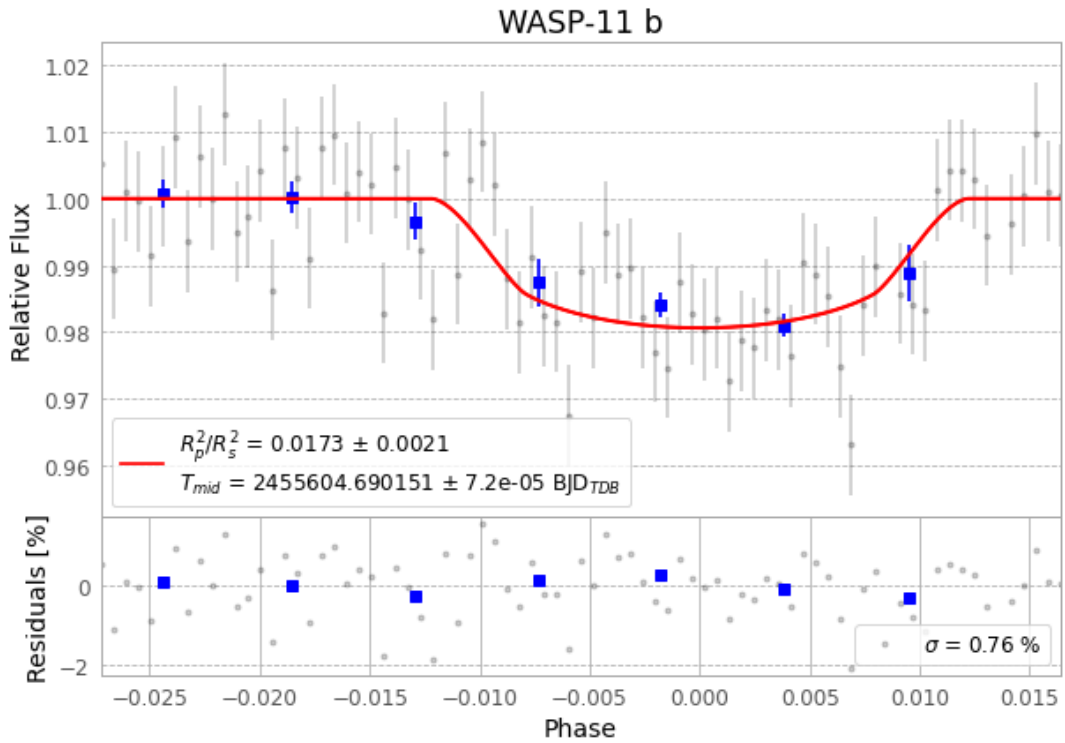


Figure 1 — FinalLightCurve_WASP-11 b_2011-02-11.png (SHA-256 6f18d5939542ddf...

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [380, 201], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 c108cc85274fb88d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

79 FITS frames (52 MB), HATP-10110212020616.FITS → HATP-10110212060013.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-10-110212.log (def4a2fea86c...), FinalLightCurve_WASP-11 b_2011-02-11.png (6f18d5939542...), FinalLightCurve_WASP-11 b_2011-02-11.csv (6d580f96c248...)

Compute resources

Wall time 172.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-10 23:40Z.